

ATS HANDLING

Kinshasa International Airport — FIH

Police Bagage

Technical documentation

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1. Overview

Police Bagage is a platform covering the full life cycle of the passenger and their baggage, from check-in at the counter to loading into the aircraft hold, including boarding at the gate. The system intercepts baggage-fraud attempts in real time and gives supervisors live monitoring and activity reports.

The platform consists of five applications and one application programming interface (API), built around a single database.

Component	Audience	Role
Mobile app	Ramp agents	Scan boarding passes and baggage on Zebra PDAs
Web dashboard	Supervisors, admins	Live monitoring, reports, account management
Disputes app	Supervisors	Handle baggage disputes and claims
Baggage tracking	Passengers (public)	Check a bag's status
Today's flights	Passengers (public)	Flight board and statuses
API	Internal apps	Scan and anti-fraud logic

2. Technical architecture

The project is organised as a monorepo. Source code is shared across applications, ensuring consistency of types and business rules.

```
/
  apps/
    mobile/    React Native (Expo) app – Zebra agents
    web/       Next.js dashboard – supervisors and admins
    litige/    Next.js disputes app – supervisors
    tracking/  Public Next.js bag tracking – passengers
    vols/      Public Next.js today's flights – passengers
  packages/
    api/       Fastify service (scan, anti-fraud)
    bcbp-parser/ Boarding pass and tag parsing
    shared/    Shared types and rules
    supabase/migrations/ Database schema
```

Technologies

Layer	Technology
Mobile app	React Native, Expo, Expo Router
Web apps	Next.js (App Router), React
API service	Node.js, Fastify
Database	Supabase (PostgreSQL)
Realtime and authentication	Supabase Realtime, Supabase Auth
Reports	ExcelJS

Access separation

- Public apps (tracking, flights) require no authentication; reads happen server-side with a service key, returning only non-sensitive data.
- Internal apps (dashboard, disputes) are protected by authentication; permissions are enforced by both the database and the interface.
- The mobile app talks to the database through the API service, which centralises the anti-fraud rules.

3. Roles and permissions

Role	Platform	Permissions
Administrator	Web	Create and manage accounts; all supervisor actions
Supervisor	Web, Disputes	Dashboard, flight statuses, reports, disputes
Agent	Mobile	Scan, load into the hold, reroute

The account management page is strictly reserved for administrators. A supervisor attempting to access it is automatically redirected to the dashboard.

4. Data model

Table	Main content
profiles	Users: name, role, assigned gate
flights	Flights: number, origin, destination, stops, times, date, status
passengers	Passengers: name, PNR, seat, class, declared bags, boarding
passenger_legs	Passenger legs (connecting flights)
baggage	Baggage: tag, serial, registered, loaded in hold, rush, timestamps
fraud_alerts	Fraud alerts: passenger, PNR, tag, reason, gate
baggage_disputes	Disputes and claims: status, reason, passenger origin

Possible flight statuses are: Scheduled, Boarding, Closed and Cancelled. On the baggage table, three indicators summarise progress: registered on the belt, loaded into the hold, and marked for rerouting.

5. Baggage life cycle

The bag goes through successive states. This sequence is central to the system and is reflected in the tracking offered to the passenger.

State	Meaning
Pending	Declared on the boarding pass, tag not yet scanned on the belt
Registered	Tag scanned on the belt, anti-fraud check passed
Loaded in hold	Bag loaded into the destination hold (Load function)
Rerouting	Remaining bag marked for the next flight (Rush function)

Recommended operational flow: after scanning bags on the belt, the agent first scans the bags to be rerouted (Rush), then triggers the bulk loading of the remaining registered bags (Load).

6. Anti-fraud rules

The check applies when a bag tag is scanned. Five rejection rules are enforced and can never be bypassed.

Detected situation	Decision	Alert
Passenger not registered for this flight	Bag rejected	Yes
Zero bags declared on the boarding pass	Bag rejected	Yes
Declared bag count exceeded	Bag rejected	Yes
Tag already scanned on this flight	Bag rejected	No (duplicate)
Bag belongs to another flight	Bag rejected	No

The baggage agent is never at fault: they scan what comes on the belt. Fraud originates at the check-in counter. The system intercepts the item before it goes into the hold and alerts the supervisor. Alerts are flagged and displayed, with no resolution workflow in the application: they serve as a record and a signal to act.

Note on the shared airline code

The numeric code 071 corresponds to both Ethiopian Airlines and Air Congo, which share the IATA code ET. This is not an anomaly. The system never relies on the airline code to link a bag to a passenger: the link key is always the serial number combined with the flight and the date.

7. Applications

Mobile app (agents)

Intended for Zebra Android PDAs, it relies on DataWedge, which injects scans as keystrokes. Flow: sign in, select the flight, then five functions — Check-in, Baggage, Boarding, Load, Rush.

Web dashboard (supervisors and administrators)

- Overview of today's flights (departures and arrivals).
- Flight detail: passengers, boarding, confirmed bags, loaded in hold, rerouting, alerts.
- Flight status change; Reports page; Accounts page reserved for administrators.

Disputes app (supervisors)

- Baggage list filterable by date, flight, loading and dispute status.
- Opening and tracking cases; receiving passenger claims; daily Excel report.

Tracking and Today's flights (passengers)

- Tracking: search by PNR (all bags) or by tag (a single bag); problem reporting.
- Today's flights: board of the day's flights with status and estimated delay; links to the airport portal.

8. Reports

Two types of Excel report are produced by the dashboard.

- Flight report: full detail of a flight in five sheets — Summary, Passengers, Baggage, Fraud alerts, and statistics for the day.
- Period report: aggregated review over the chosen period (day, week, month, year or custom range) in five sheets. The accounting summary presents volumes, averages and rates.

9. Deployment

Web applications

The four web applications are deployed on Hostinger Cloud from independent GitHub repositories, with automatic deployment on the main branch (Next.js framework, npm package manager, Node 20 or later).

Application	Repository	Local port
Dashboard	police-web	3000
Baggage tracking	police-tracking	3002
Disputes	police-litige	3003
Today's flights	police-vols	3004

Environment variables (Supabase address and keys, hub) are provided in the host configuration and are never written into the code. HTML pages are served without caching so they always reference the current version; static assets keep a long cache, and a recovery mechanism reloads the page in case of a stale resource after an update.

API service and mobile app

The API service is deployed separately on Hostinger Cloud from a standalone repository. The mobile app is built with Expo EAS and published on Google Play; the internal version number increments automatically on each build.

10. Security and privacy

- Communications are encrypted end to end.
- Sensitive keys remain server-side and are never exposed to clients or written into the code.
- Public apps collect no personal data for commercial purposes and include no advertising trackers.
- Access to internal data is limited to authorised accounts.
- Anti-fraud rules cannot be bypassed by any role; any exception is a manual supervisor action.

ATS Handling — Kinshasa International Airport (FIH). Internal reference document.